Project Title

Team members name

Branch

Course

Mentor name

Company name

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Outline

1. Introduction
2. Scope and Applications
3. Literature Survey
4. Comparison
5. Gap Identified
6. Problem Statement
7. Objectives
8. Proposed System
9. Design
   - Block Diagram
10. Dataset
11. Conclusion
12. References

Introduction

- 'Sea Level Rise Prediction'is essential for protecting coastal areas from the risks of rising water levels caused by climate change.
- Accurate sea level predictions are essential for climate impact assessment and infrastructure planning.
- Traditional forecasting methods struggle with the complexities of time-series data, which include both seasonal variations and long-term trends.
- To improve accuracy, advanced techniques integrate temporal and spatial data, addressing the uncertainties in forecasts.
- Offers faster and more precise predictions while accounting for uncertainties inherent in sea level rise forecasts.
- Highlights the ability to measure and address prediction uncertainties, essential for robust decision-making in risk-prone areas.

Scope and Applications

Scope:

- Improved prediction accuracy
- Model generalization for global coastal areas
- Real-time sea-level monitoring
- Multidisciplinary impact (climate science, urban planning)
- Technological advancements in geospatial data analysis

Applications:

- Coastal risk management
- Urban planning and infrastructure development
- Environmental monitoring
- Policy-making and governance and real estate risk assessment

A machine learning approach to evaluate coastal risks related to extreme weather events in the Veneto region (Italy)

Maria Katherina Dal Barco, Margherita Maraschin,Davide Mauro Ferrario

- The study examines how machine learning (ML) is used to assess the impacts of extreme weather on coastal areas in the Veneto region, Italy.
- The sources of the dataset are ARPAV (Agenzia Regionale per la Prevenzione e Protezione Ambientale del Veneto), JRC (Joint Research Centre), and various coastal monitoring stations.
- The study uses datasets on land use, population, soil, topography, weather, and sea level to train models predicting extreme weather impacts in Veneto.The dataset size is approximately 10 MB.
- The study used three models:
  1. Multi-Layer Perceptron (MLP)
  2. Random Forest (RF)
  3. Support Vector Classifier (SVC)

Models used and Results:

- MLP was the most efficient, similar to SVC, while RF had slightly lower performance based on sea surface height, precipitation, and wind intensity.
- Evaluation metrics include precision, recall, F1 score, and weighted metrics. Features include atmospheric, oceanographic, temporal, and geographical factors. The dataset was cleaned to address missing data and errors before analysis.
- MLP and SVC had similar F1 scores (0.94 and 0.95), while RF scored 0.92. All models had high recall (over 0.91) but low precision (0.12-0.14), indicating many false positives.
- The models effectively identified factors contributing to extreme weather impacts in the Veneto region, though improving precision is still a challenge.

Advantages:

- High Performance: MLP achieved a weighted F1 score of 0.94, showing strong risk estimation.
- Feature Importance: The model highlights key hazard factors, supporting targeted adaptation strategies.
- Robustness and Explainability: The algorithm ensures clear, reproducible, and reliable results.
- Multi-Risk Assessment: Combines hazard, vulnerability, and exposure factors for a complete view of coastal risks.

Disadvanatges :

- Impact data is limited and sometimes inaccurate, reducing reliability.
- Missing details about economic impacts and event locations make assessments harder.
- Results can be complex and difficult for non-experts to understand

Water Level Prediction Model Applying a Long Short-Term Memory (LSTM)–Gated Recurrent Unit (GRU) Method for Flood Prediction

Minwoo Cho , Changsu Kim , Kwanyoung Jung and Hoekyung Jung

- Flooding, caused by heavy rainfall, requires accurate water level prediction models.
- This study uses a hybrid Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) model for time-series prediction.
- The dataset contains 76,152 rows of water level and meteorological data from October 2013 to June 2022, with 75percent for training and 25percent for testing. - Sourced from WAMIS and the Korea Meteorological Administration, the dataset includes data from AWS(Automaic weather Division) and ASOS(Automated Synoptic Observing System) stations, estimated at 10-50 MB in CSV format.

Models used and Results:

- The study used Multi-LSTM, Multi-GRU, and hybrid LSTM-GRU models, with the LSTM-GRU model performing best across nine scenarios.
- It showed the highest accuracy in predicting high water levels, particularly when combining water level data with the ASOS dataset for flood management.
- Performance was evaluated using MSE, NSE, and MAE, with the LSTM-GRU model achieving an NSE of 0.920 in Scenario 1 (using only water level data) and 0.942 in Scenario 3 (combining water level and ASOS data). Scenario 2, combining water level and AWS data, also showed strong performance with an NSE of 0.920.

Advantages :

- Simplifies predictions using meteorological and water level data
- Improving accuracy, especially with ASOS data, and providing insights into LSTM and GRU models' effectiveness for flood management.

Disadvantages :

- Performance may vary with observation point distance, limiting applicability.
- Reliance on specific datasets reduces versatility, and outliers may impact prediction accuracy.

Sea Level Prediction Using Machine Learning

Rifat Tur, Erkin Tas, Ali Torabi Haghighi and Ali Danandeh Mehr

- Accurate sea level prediction is vital for coastal management, especially in areas like Antalya Harbor, Turkey.This study uses machine learning models, including MLR(Multi Linear Regression) and ANFIS(Adaptive Neuro Fuzzy Interference System), to improve forecasts and account for key weather factors amid rising sea levels from climate change and human activities.
- The Turkish National Sea Level Monitoring System (TUDES) now operates 20 active stations that measure sea level and meteorological factors like air temperature and pressure.
- The study used correlation analyses to identify how sea level relates to variables like air temperature and pressure over time.
- Based on these relationships, lagged values of sea level, air temperature, and air pressure were selected as key inputs for the prediction models.

Models used and Results:

- MLR establishes a linear relationship between sea level and factors like lagged values and meteorological data, minimizing prediction errors.
- ANFIS combines neural networks and fuzzy logic, using IF-THEN rules to capture non-linear relationships and improve accuracy during training.
- ANFIS outperformed MLR in sea level predictions with lower error values (RMSE, MAE) and higher Nash-Sutcliffe Efficiency (NSE) scores.
- ANFIS showed better performance for 24-hour forecasts.
- ANFIS is more effective for sea level prediction, especially when considering meteorological factors, improving forecasting accuracy for coastal management.

Advantages :

- Higher Accuracy: ANFIS provides higher accuracy with lower error values compared to MLR.
- Non-Linear Relationships: ANFIS captures complex non-linear relationships that MLR may miss.
- Adaptability: ANFIS adapts predictions using neural network learning and fuzzy logic, making it more robust in varying conditions.

Disadvantages :

- Limited Applicability: Findings are specific to the case study area, limiting their use in other locations.
- Data Scarcity: Long-term sea level data, crucial for reliable predictions, may be scarce due to high monitoring costs.
- Narrow Focus: The study mainly focused on oceanographic and meteorological factors, neglecting other hydrological processes like rainfall and runoff.

Indian Ocean Dynamic Sea Level, Variability And Projections In CMIP6 Models

Georgios Boumis, Hamed R. Moftakhari, and Hamid Moradkhani

- This study focuses on simulating historical sea level changes and variability in the Indian Ocean using outputs from 27 CMIP6 models.
- It analyzes historical simulations spanning 1850 to 2014 and explores future projections under two Shared Socioeconomic Pathways (SSP) scenarios.
- The datasets include CMIP6, or the Coupled Model Intercomparison Project Phase 6, is an international effort to improve our understanding of the Earth's climate system and predict future climate changes model outputs sourced from the ESGF(Earth System Grid Federation) portal and wind field data from the ECMWF ERA5 (European Centre for Medium-Range Weather Forecasts ERA5: ECMWF Reanalysis 5th Generation)dataset.
- The models feature varying grid resolutions, such as 360 x 300 and 1051 x 1442, providing a comprehensive analysis of sea level dynamics in the region.

Models used and Results:

- CMIP6 models were evaluated using statistical methods like RMSE and skill scores.
- Only 7 out of 27 models met benchmarks for sea level and wind accuracy.
- EC-Earth3 (European collaborative Earth System Model)and NorESM2-MM(Norwegian Earth System Model) were rejected for unrealistic results.
- Reliable models like BCC-ESM-MR(BCC: Beijing Climate Center ESM: Earth System Model)and CNRM-CM6-1(CNRM: Centre National de Recherches Météorologiques (French National Center for Meteorological Research). were chosen.
- Improved accuracy and reduced uncertainty in future sea level and wind predictions.

Advantages:

- Improved Accuracy
- Reduced Uncertainty: Excluding poor models improves forecast reliability.
- Regional Insights:CMIP6 enables detailed analysis of sea level changes.

Disadvanatges :

- Model Gaps: Some models may miss key climate processes.
- Decision Risks: Excluding models may lead to incomplete data use.
- High Costs: Mitigation strategies could become expensive.

Uncertainty Analysis of Numerical Simulation of Seawater Intrusion Using Deep Learning-Based Surrogate Model

Tiansheng Miao, He Huang, Jiayuan Guo, Guanghua Li, Yu Zhang and Naijia Chen

- Predicting seawater intrusion is difficult for traditional models due to rising sea levels and random changes.
- The dataset includes 100 training and 20 validation sample groups, with inputs like sea level rise and well pumping capacities.
- The dataset used in the study likely comes from various sources, including groundwater monitoring networks such as the U.S. Geological Survey (USGS), the European Environmental Agency (EEA), or India's Central Ground Water Board (CGWB)
- Three models—DBNN(Deep Belief Neural Network), RBF(Radial Basis Function Network), and DCNN(Deep Convolutional Neural Network)—were used for seawater intrusion predictions, with DBNN showing the highest accuracy

.

Models used and Results:

- Uses DBNN and DCNN to create surrogate models for seawater intrusion predictions, reducing computation time in Monte Carlo simulations.
- DBNN maps input-output relationships, while DCNN extracts features through convolution layers.
- Accurately predicted chloride ion concentrations and seawater intrusion areas, showing varying risks across wells.
- The average seawater intrusion area was 69.39 km^2, close to the deterministic model's prediction of 68.5 km^2.
- Standard deviation of 7.31 km^2 showed that sea level rise had less impact on intrusion than chloride levels.
- The study used Monte Carlo simulations to analyze uncertainty in seawater intrusion due to sea level rise.

Advantages:

- Better Accuracy: Deep learning models improve seawater intrusion predictions.
- Uncertainty Analysis: Monte Carlo method helps understand prediction variability.
- Practical Insights: Offers solutions for groundwater management.
- Efficient: Reduces computation time without losing accuracy.

Disadvanatges :

- Limited Data: Few training samples may affect model reliability.
- Regional Focus: Findings may not apply to other areas.
- Oversimplification: Treats sea level rise as random, ignoring complex factors.
- High Initial Costs: Setup and validation are resource-intensive.
- Short-term Data: Lacks long-term seawater intrusion trends.

Predicting Sea-Level Variations in the Northwestern Arabian Gulf Using Advanced Statistical and Deep Learning Model

Abather Jabbar Bashar Alhallafa, Javier Vilcáeza

- This research studies sea-level fluctuations in the northwestern Arabian Gulf, focusing on meteorological factors like temperature, pressure, wind speed, and direction.
- The study combines traditional forecasting methods with machine learning to improve sea-level prediction accuracy.
- The goal is better disaster preparedness and coastal management, addressing climate change and rising sea levels.
- The dataset is obtained from an offshore measuring station near Faw Grand Port in the marine waters of Iraq, managed by General Acoustics.
- The dataset comprises 16,092 data points, covering meteorological and environmental measurements from 2016 to 2022.

Models used and Results:

- The study uses SARIMA, SARIMAX, CNN, LSTM, and CNN-LSTM models to predict sea-level variations based on meteorological data from 2016 to 2022.
- The aim is to understand how factors like temperature, pressure, and wind influence sea levels for climate and hydrological analysis.
- SARIMA (Seasonal AutoRegressive Integrated Moving Average)and SARIMAX (Seasonal AutoRegressive Integrated Moving Average with Exogenous variables)focus on time-series forecasting, with SARIMAX adding meteorological data.
- CNN captures spatial patterns, while LSTM handles sequential data.
- The combined CNN-LSTM model integrates both spatial and temporal features for more accurate predictions.
- The LSTM model outperformed others with the lowest Mean Squared Error (0.02425) and Root Mean Squared Error (0.15574), proving effective for sea-level predictions.

Advantages:

- Improved accuracy with LSTM helps coastal communities make better decisions.
- Diverse meteorological data provides a comprehensive understanding of sea-level changes.
- Insights support informed policies and disaster plans for climate change adaptation.

Disadvanatges :

- SARIMAX model showed lower accuracy, requiring better techniques.
- Predictions are limited by the quality of available meteorological data.
- Seasonal changes and interactions may affect long-term forecast precision.

An interpretable hybrid deep learning model for flood forecasting based on Transformer and LSTM

Wenzhong Li , Chengshuai Liu , Yingying Xu , Chaojie Niu , Runxi Li ,Ming Li, Caihong Hu, Lu Tian

- Flood forecasting is challenging due to complex factors.
- The AGRS(Attention-Gated Recurrent Skip Long Short-Term Memory)-LSTM-Transformer hybrid model combines advanced algorithms to improve prediction accuracy and interpretability, enhancing early warning and water resource management.
- The study used hydro-meteorological data, including runoff and rainfall data from Jingle watershed, 14 stations, covering 98 flood events from 1971 to 2013 at 1-hour intervals.The size of dataset be around 10-20 MB.
- Hydro-meteorological data, including runoff and precipitation, is normalized and structured for model input.

Models used and Results:

- The AGRS-LSTM-Transformer model combines Long Short-Term Memory (LSTM) networks with Transformer architecture to improve flood forecasting accuracy by processing key flood variables.
- Combines LSTM for time sequences, Transformer for global dependencies, and convolution layers for feature extraction, with dropout to prevent overfitting.
- Hyperparameter Optimization: AGRS optimizes based on Nash Sutcliffe Efficiency (NSE) metrics.
- Achieved NSE > 0.905, outperforming traditional models in flood forecasting, particularly for temporal features.
- Flood Forecasting: Excellent performance with NSE > 0.905 for 1-6 hour lead times, accurately predicting runoff using rainfall data and nearby stations.

Advantages :

- Enhanced Accuracy: AGRS-LSTM-Transformer improves flood forecasting with higher NSE values, outperforming traditional models.
- Interpretability: SHAP values explain the model's predictions, improving understanding.
- Adaptability: AGRS optimizes hyperparameters dynamically, boosting performance.

Disadvantages :

- Data Confidentiality: Confidential data limits result validation and reproducibility.
- Model Complexity: The hybrid model's complexity and high computational needs may hinder practical use.
- Generalizability: The model's effectiveness in other regions or climates needs further testing.

Flood prediction with time series data mining: Systematic review

Dimara Kusuma Hakim, Rahmat Gernowo, Anang Widhi Nirwansyah

- Flood disasters are worsening due to climate change and urbanization. Current models struggle with non-linear data and overfitting.
- Flood prediction research is most active in China and India, followed by Iran and the United States, with Southern Asia leading in Asia, followed by Eastern and South-Eastern Asia.
- The datasets used include satellite data (e.g., from Sentinel and Landsat), geospatial data, and hydrological station data with size of 1 GB or more.
- This system combines Genetic Algorithms (GA) and Adaptive Neuro-Fuzzy Inference Systems (ANFIS) for more accurate flood predictions and better disaster management.
- The study uses historical flood data from meteorological records, river discharge, remote sensing, and factors like precipitation, soil moisture, land use, and past floods.

Models used and Results:

- Models implemented include Artificial Neural Networks (ANN) for non-linear patterns, Support Vector Regression (SVR) for robust regression, and Long Short-Term Memory (LSTM) for time series forecasting. Hybrid models are also employed to enhance predictive accuracy by combining multiple algorithms.
- Hybrid models: Combine data mining (neural networks, decision trees) with optimization (genetic algorithms) for better accuracy.
- Time series models: Use historical data and Ant Colony Optimization for forecasting.
- Hybrid models improve accuracy, with higher R-squared and lower MAE.
- Effective for real-time flood forecasting and disaster management.

Advantages:

- Advanced algorithms (e.g., Genetic Algorithms, ANFIS) improve prediction accuracy over traditional models.
- Effectively handle non-linear data and diverse meteorological/hydrological datasets for reliable forecasts.

Disadvanatges :

- Dependence on high-quality historical data, which may be unavailable.
- High computational requirements limit real-time application.
- Limited generalizability to regions with different hydrological conditions.

A deep learning approach to predict significant wave height using long short-term memory

Felipe C. Minuzzi a,Ocean Modelling

- Accurate wave height predictions are crucial for navigation, renewable energy, and ocean engineering.
- This paper develop a novel LSTM training methodology for forecasting significant wave heights along Brazil's coast.
- Use historical wave data, focusing on lead time preservation, with univariate and multivariate forecasting methods.
- The dataset used is the ERA5 hourly data from the Copernicus Climate Change Service (C3S), covering the time span from 1979 to 2018 approximately 10-15 GB.
- This paper enhance wave height prediction reliability, supporting diverse maritime activities.

Models used and Results:

- Features and Data: Extracted significant wave height, peak wave period, and 10m wind speed using data from Rio Grande buoy and ERA5 dataset.
- Data Split: 20 percentage for validation, remaining for testing predictions.
- LSTM neural network with three layers (64, 48, 32 neurons), optimized using MAE loss function and Adam algorithm, with early stopping applied.
- The LSTM model performed well in predicting wave heights.

Advantages:

- High accuracy: Reaches up to 97.25 percentage accuracy for certain lead times.
- Computational efficiency: Requires fewer resources compared to traditional models.
- Adaptive: Effectively utilizes historical data for improved predictions.

Disadvanatges :

- Lack of interpretability: Operates as a "black box," making its decision-making process unclear.
- Data dependency: Relies heavily on the quality of training data.
- Reduced long-term accuracy: Forecasting accuracy decreases as lead times increase.

Optimized Deep Learning Model for Flood Detection Using Satellite Images

Andrzej Stateczny , Hirald Dwaraka Praveena, Ravikiran Hassan Krishnappa,Kanegonda Ravi Chythanyaand Beenarani Balakrishnan Babysaroja

- Urban areas face heightened flood risks, with traditional detection methods lacking accuracy and timeliness.
- A Deep Hybrid Model for Flood Prediction (DHMFP) uses satellite imagery for improved accuracy.
- Combines k-means clustering for segmentation with CNN and deep ResNet for classification, optimized by the CHHSSO(himera Hybrid Harmony Search Social Optimization). algorithm.
- Dataset: Includes Landsat 8 (Feb 2018) and Sentinel-2 (Aug 2018) satellite images from the Kerala floods. The dataset is taken from the NASA Earth Observatory website, specifically the page titled "Before and After the Kerala Floods."
- Features includes 13 spectral bands and indices like Difference Vegetation Index,Normalized Difference Vegetation Index (NDVI),combined into a feature set (Fset) for hybrid model classification.

Models used and Results:

- DHMFP combines CNN and ResNet, optimized with CHHSSO, for flood prediction using satellite images.
- Preprocessing includes Median filtering and cubic chaotic map weighted k-means clustering identify flood areas.
- Achieved 95.21 percentage accuracy, 93.48 percentage sensitivity, and 98.29 percentage specificity.
- Error Rates: False Negative Rate (FNR): 0.0858; False Positive Rate (FPR): 0.024.
- The DHMFP model improves flood prediction accuracy using deep learning and remote sensing.

Advantages:

- Efficiency: Automatically extracts features from raw data, streamlining training.
- Accuracy: Utilizes advanced deep learning for precise flood prediction.
- Scalability: Leverages remote sensing for large-scale flood mapping without detailed inputs.

Disadvanatges :

- Complexity.
- Overfitting: ResNet's complexity may cause overfitting, reducing interpretability.
- Urban Challenges: Mapping floods in urban areas is difficult due to their complexity.
- SAR Limitations: Shadowing effects in SAR data can affect urban flood detection accuracy.

Leveraging a Novel Hybrid Ensemble and Optimal Interpolation Approach for Enhanced Streamflow and Flood Prediction

Mohamad El Gharamti, Arezoo RafieeiNasab, and James L. McCreight

- Flooding is increasing, requiring better forecasting. The paper proposed EnKF-OI(Ensemble Kalman Filter and Optimal Interpolation) hybrid model, integrated with the HydroDART system, enhances streamflow predictions by refining initial conditions and using real-time data more effectively.
- Dataset used include 42-year data from the National Water Model (NWMv2.1) used for the hybrid filter.
- Key Inputs taken includes Streamflow, soil moisture, and snow data are included for flood forecasting.
- Observational Data: Satellite and in-situ data are incorporated to improve model accuracy and performance.

Models used and Results:

- This study uses the hybrid Ensemble Kalman Filter-Optimal Interpolation (EnKF-OI) model within the HydroDART system, built on the Weather Research Forcasting-Hydro framework
- The model generates multiple streamflow predictions by running the hydrologic model until new data is available.
- Model Update: Bayes' rule updates streamflow and groundwater variables.
- Adjusts initial conditions, reduces biases using USGS(United States Geological Survey) streamflow data, and incorporates random noise for ensemble forecasting.
- The hybrid EnKF-OI model achieved NSE(Nash-Sutcliffe Efficiency) of 0.97 and KGE(Kling-Gupta Efficiency) of 0.94, improving streamflow forecasts.
- Improvement: Reduced RMSE to 78.73 and lowered forecast uncertainty by assimilating observational data.

Advantages:

- Improves flood forecast accuracy, especially for peak flows.
- Reliable in extreme flooding events
- Uses multiple data sources for better predictions

Disadvanatges :

- Requires high computational resources and expertise.
- Needs careful calibration for different areas.
- Less effective in regions with limited data.

Shoreline changes over last five decades and predictions for 2032 and 2042: a case study of Kakinada coast, Southeast coast of India

Bejjam Vaila1,. Sudhakar Ilango1

- The study analyzes shoreline changes along Kakinada coast (1972–2022) and predicts conditions for 2032 and 2042, focusing on erosion, accretion, and vulnerable areas to guide coastal management.
- Using Landsat satellite imagery from USGS Earth Explorer, shoreline attributes like DSAS data and shape length were analyzed with statistical methods, including Endpoint Rate (EPR), Linear Regression Rate (LRR), and Net Shoreline Movement (NSM).

Models used and Results:

- The Digital Shoreline Analysis System (DSAS) was used to calculate shoreline change metrics like EPR(Efficiency Performance Ratio), LRR(Logistic Regression Rate), and NSM(Normalized Streamflow Mode), analyzing historical data to quantify erosion and accretion rates.
- The Kalman Filter model predicted future shoreline conditions, enhancing accuracy for 2032 and 2042 forecasts.
- The study found a maximum erosion rate of -31.27 m/year and maximum accretion of 22.74 m/year.
- By 2042, the Kakinada coast is expected to experience 891 m of erosion and 700 m of accretion. Vulnerable areas were identified for targeted coastal management.

Advantages:

- Comprehensive Analysis: Detailed six-decade shoreline change insights.
- Predictive Capability: Kalman filter model allows accurate future forecasts.
- Targeted Management: Identifies areas for focused coastal protection.

Disadvanatges :

- Data Quality: Predictions may have uncertainties due to data and model limitations.
- Historical Dependence: Reliance on past data may miss future changes.
- Complexity: Coastal dynamics and human activities complicate predictions.

Ocean Wave Height Forecasting using Deep Learning Neural Networks and Optimization Techniques

M.Lohitha Chowdary,P.Premsai,M. Dasharna,I.Sri Kavya

- The study aims to improve ocean wave height predictions, crucial for maritime safety, by addressing the limitations of traditional models.
- The study uses the Mooloolaba Beach dataset, consisting of 16,003 wave records sampled every 30 minutes over a year.
- Key features include wave height, period, and other wave characteristics. The data is split into 80 percentage for training and 20 percentage for testing to assess model performance

Models used and Results:

- It combines an Artificial Neural Network (ANN) with an ANOVA Basis Kernel and uses an ensemble approach with Multi Additive Regression Trees (MART).
- The study used single-shot and multi-shot LSTM, CNN-LSTM hybrid, Bidirectional LSTM, and ANN with ANOVA Basis Kernel to predict ocean wave heights.
- SOS and MFO optimization techniques were applied to improve performance and reduce overfitting.
- The ANN with the ANOVA kernel had the highest accuracy (94.5 percent) and low RMSE (0.0469), outperforming LSTM models (CNN-LSTM: 82.57percent, LSTM: 82.15percent, multi-shot LSTM: 77.69percent).

Advantages:

- The ANN with an ANOVA Basis Kernel achieved 94.5 percent accuracy, outperforming traditional models.
- Advanced optimization improved performance and reduced overfitting.
- A large dataset (16,003 samples) ensured robust training and generalization.

Disadvanatges :

- Hybrid models are computationally intensive and require expertise.
- Model effectiveness may vary with different datasets and locations, and accuracy depends on data quality.

Deep Learning of Sea-Level Variability and Flood for Coastal City Resilience

Omid Memarian Sorkhabi, Behnaz Shadmanfar , Mohammed M.

- This study aims to explore the relationship between floods and sea-level rise in coastal cities, focusing on Gothenburg, to improve resilience against climate change.
- Coastal areas face higher flooding risks and infrastructure damage from climate change, with traditional methods failing to capture complex environmental interactions.
- The study uses a dataset combining tabular data (e.g., sea surface temperature, wind speed) and Landsat images to predict sea-level changes and assess flood impacts.

Models used and Results:

- Combines CNN and LSTM networks to predict sea-level variability.
- CNN layers extract features from data (e.g., sea level, temperature), and LSTM layers capture long-term dependencies.
- Dropout is used to prevent overfitting. Hyperparameters are optimized using the Adam optimizer (learning rate 0.01) over 500 epochs.
- The model is trained on 80 percent of the data, with 20 percent reserved for validation.
- Findings: The model predicts a potential sea-level rise of 5-10 meters in Gothenburg, affecting over 50 percent of the city in worst-case scenarios.
- It confirms increased flood frequency and magnitude under the RCP 8.5 climate scenario, highlighting the need for improved coastal resilience strategies.

Advantages:

- The deep learning model accurately captures complex non-linear relationships in environmental data, providing reliable sea-level predictions.
- It uses extensive satellite and ground sensor data for better accuracy, supporting urban planning and decision-making.
- The model also simulates flood scenarios, helping identify vulnerable areas for disaster management.

Disadvanatges :

- Requires large datasets for optimal accuracy, which may not be available in all regions.
- Complex model interpretation is challenging for non-experts, and there's a risk of overfitting with limited data.

Comparison Table

Paper	Dataset	Methods	Advantages	Disadvantages
[1]	Dataset, sourced from ARPAV, JRC, and coastal monitoring stations, includes land use, population, soil, topography, weather	MLP,RF, and SVC.MLP with high accuracy F1 Score 0.94.	High Performance,Robustness and Explainability.	Data Quality Issues,Struggles with precision and model complexity
[2]	The dataset, sourced from WAMIS and KMA, contains 76,152 rows of water level and meteorological data (Oct 2013–Jun 2022)	Hybrid model LSTM and GRU ,NSE of 0.920 and a maximum water level error of 79.64 cm in S1	Improving accuracy	Limiting applicability. Reliance on specific datasets reduces versatility
[3]	TUDES operates 20 stations monitoring sea level and meteorological data, including air temperature and pressure.	MLR and ANFIS ,ANFIS provides higher accuracy with lower error values compared to MLR.	Higher Accuracy,Adaptability	Limited Applicability,Data Scarcity.

Paper	Dataset	Method	Advantages	Disadvantages
[4]	CMIP6 models were evaluated using statistical methods like RMSE and skill scores.	Three models—DBNN, RBF, and DCNN—were used for seawater intrusion predictions,	Improved Accuracy Reduced Uncertainty	generalized assumptions,Dependent on past data.
[5]	The dataset includes 100 training and 20 validation samples, with inputs like sea level rise and well pumping, sourced from networks like USGS, EEA, and CGWB.	Three models—DBNN, RBF, and DCNN—were used for seawater intrusion predictions,	Better Accuracy,Uncertainty Analysis	Limited Data,Regional Focus,High Initial Costs.
[6]	The dataset from an offshore station near Faw Grand Port, Iraq, includes 16,092 meteorological and environmental data points (2016–2022).	SARIMA, SARIMAX, CNN- LSTM.The LSTM model excelled prediction with the lowest MSE (0.02425) and RMSE (0.15574).	Diverse meteorological data aids in understanding sea-level changes, supporting climate adaptation policies and disaster plans.	Predictions are limited

Paper	Dataset	Method	Advantages	Disadvantages
[7]	10-20 MB of hydro-meteorological data from 14 Jingle watershed stations, covering 98 flood events (1971-2013) at 1-hour intervals.	The AGRS-LSTM-Transformer model combines LSTM networks with Transformer architecture,Achieved NSE > 0.905	Enhanced Accuracy, Interpretability,Adaptability	Data Confidentiality,Model Complexity
[?]	satellite data (e.g., from Sentinel and Landsat), geospatial data, and hydrological station data with size of 1 GB or more.	hybrid models like GA)and Adaptive Neuro-Fuzzy Inference Systems (ANFIS) with higher R-squared and lower MAE.	Enhance prediction accuracy by effectively handling non-linear and diverse meteorological or hydrological data.	Dependence on high-quality historical data,Generalization
[8]	The dataset is ERA5 hourly data (C3S), spanning 1979–2018, around 10–15 GB.	A 3-layer LSTM (64, 48, 32 neurons) optimized with MAE and Adam performed well in predicting wave heights.	Computational efficiency,Adaptive	Data dependency,Reduced long-term accuracy.

Paper	Dataset	Method	Advantages	Disadvantages
[9]	Includes Landsat 8 (Feb 2018) and Sentinel-2 (Aug 2018) images of Kerala floods from NASA Earth Observatory.	hybrid CNN-ResNet model using satelite images,Achieved 95.21 percentage accuracy	Efficiency,Accuracy: Utilizes advanced deep learning for precise flood prediction.	Complexity,Overfitting, leveraging remote sensing for flood detection
[10]	Dataset used include 42-year data from the National Water Model (NWMv2.1)	Ensemble Kalman Filter-Optimal Interpolation (EnKF-OI) model within the HydroDART system,NSE of 0.97	Improves flood forecast accuracy, Reliable in extreme flooding events	Needs high computational power and careful tuning,limited data.
[11]	Landsat satellite imagery from USGS Earth Explorer	Digital Shoreline Analysis System (DSAS) for shoreline change metrics and the Kalman Filter predicts 891 m of erosion and 700 m of accretion by 2042.	Predictive Capability: Kalman filter model allows accurate future forecasts. Targeted Management	Limited by historical data,long-term view of coastal change

Paper	Dataset	Method	Advantages	Disadvantages
[12]	Mooloolaba Beach dataset, consisting of 16,003 wave records sampled every 30 minutes over a year.	The study used single-shot and multi-shot LSTM, CNN-LSTM hybrid, Bidirectional LSTM, and ANN with ANOVA Basis Kernel to predict ocean wave heights	The ANN with the ANOVA kernel had the highest accuracy (94.5 percent) and low RMSE (0.0469)	More location-specific and deals with simpler environmental factors
[13]	The study uses a dataset combining tabular data (e.g., sea surface temperature, wind speed) and Landsat images to predict sea-level changes and assess flood impacts	Combines CNN and LSTM networks to predict sea-level variability.The model predicts a potential sea-level rise of 5-10 meters in Gothenburg, affecting over 50 percent of the city in worst-case scenarios	The deep learning model predicts sea levels accurately using satellite and sensor data, aiding urban planning, flood simulation, and disaster management.	Requires large datasets,Complex model interpretation

Gap Identified

The research papers highlight several limitations in using machine learning for coastal risk assessment and flood prediction. Key challenges include data quality issues, lack of model interpretability, uncertainty quantification, and real-time application difficulties. Addressing these limitations is crucial for improving the accuracy and reliability of machine learning-based solutions for coastal risk management and flood prediction.

Problem Statement

The accurate prediction of sea level rise is crucial for coastal management and climate adaptation. However, existing models often struggle to capture the complex and dynamic patterns of sea level variations. This project aims to address this challenge by leveraging a hybrid approach that combines the strengths of time-series analysis and spatial feature extraction from satellite imagery.

Objectives

- Create a robust model combining time-series data and satellite imagery.
- Make the model's decision-making process more transparent.
- Identify vulnerable coastal regions.
- Provide Practical Insights: Offer actionable insights for coastal management.

Proposed System

- The proposed system for sea level rise prediction integrates both time-series data and satellite imagery to enhance forecasting accuracy.
- It preprocesses and normalizes these datasets, with Convolutional Neural Networks (CNNs) extracting spatial features from satellite images, such as sea surface temperature and coastal topography, while Recurrent Neural Networks (LSTM or GRU) capture temporal dependencies from historical sea level data.
- These features are combined into a single feature vector, which is split into training, validation, and test sets.
- By training the model on historical data and evaluating it using metrics like Adam optimizer, Mean Squared Error (MSE), the system provides valuable insights for coastal resilience and planning.

Design

Design

- The model design for sea level prediction combines Long Short-Term Memory (LSTM) GRU and CNN to improve forecasting accuracy.
- This sea level rise prediction model that integrates CSV time-series data and satellite imagery.
- Data is first collected and preprocessed (normalized and cleaned), followed by feature extraction where CNNs capture spatial features and LSTMs/GRUs capture temporal patterns.
- These features are then fused, and a fully connected layer makes predictions on sea level rise.
- The model is trained using MSE loss and the Adam optimizer to improve its accuracy.
- Finally, the model generates sea-level rise predictions, providing insights for coastal risk management and planning.

Block Diagram

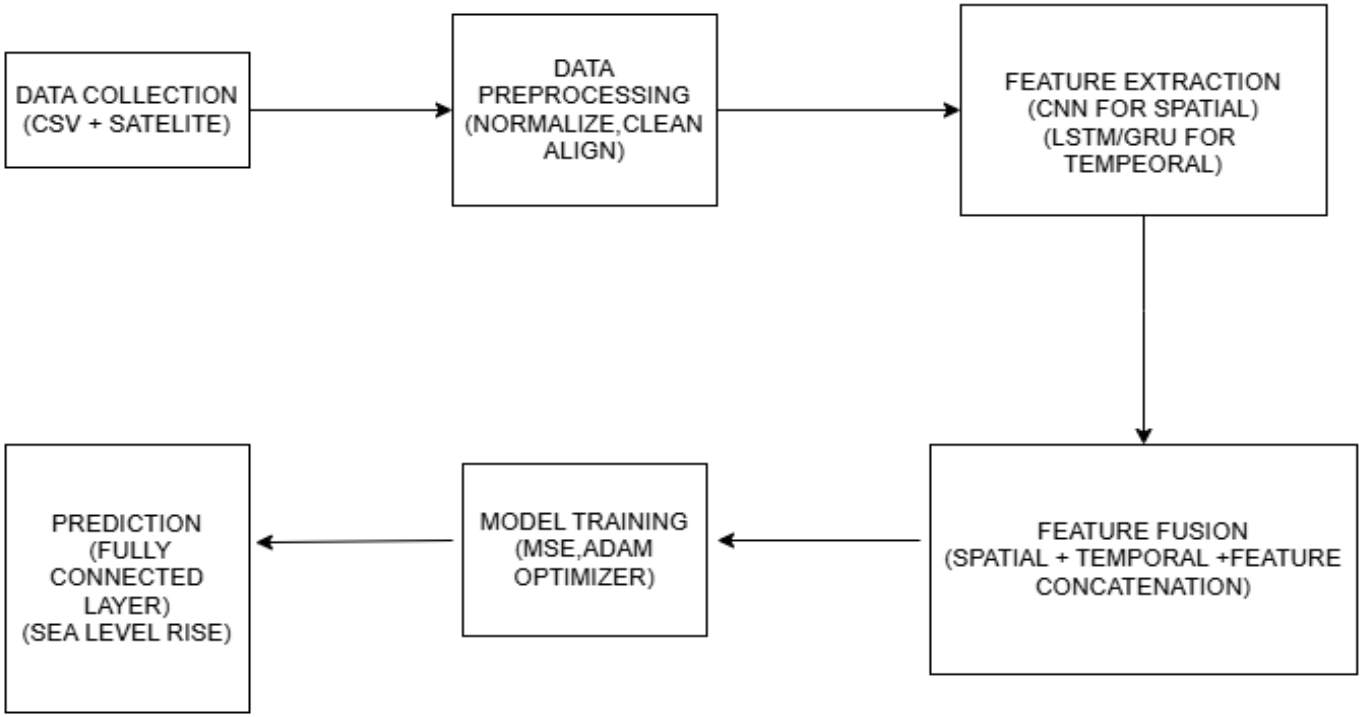


Figure: Block Diagram

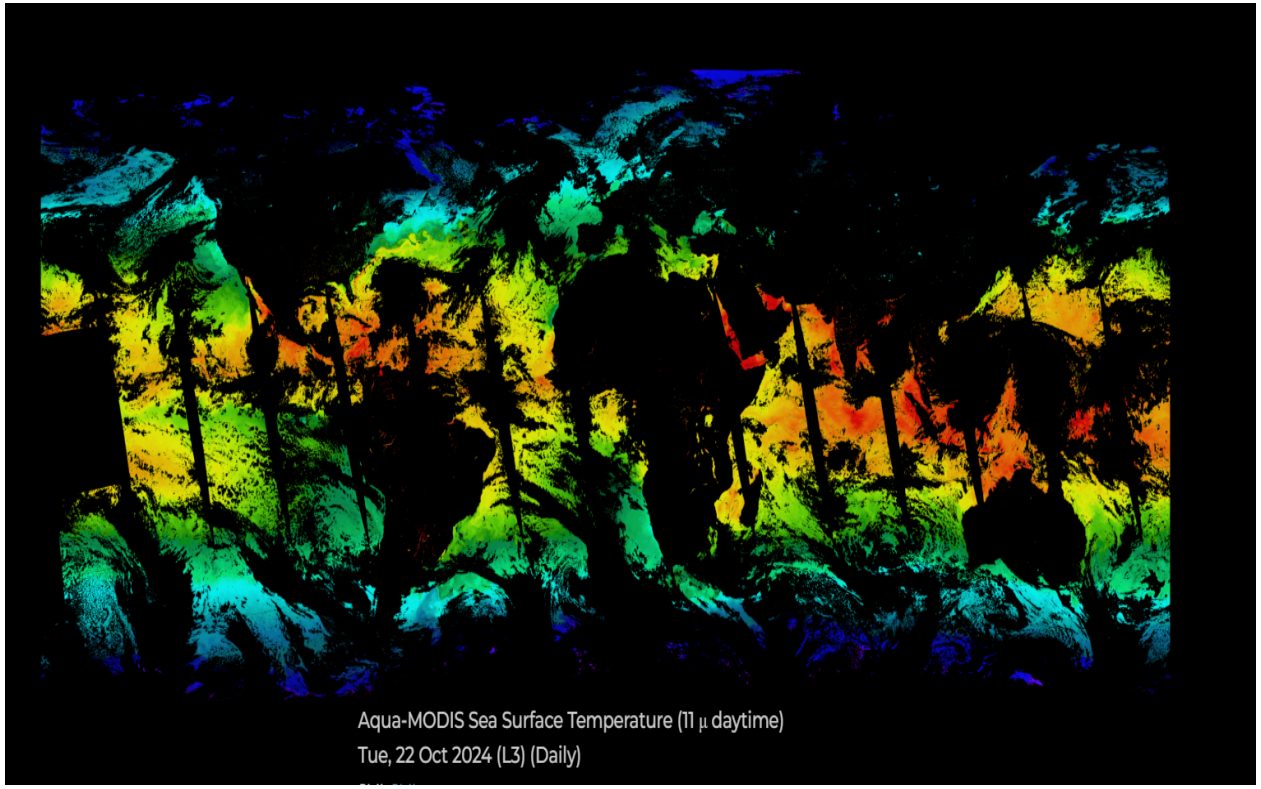
Dataset

- The dataset contains 8,323 entries with four columns, likely representing time-series sea level measurements: Year, Month, Day, and Sea Level. The dataset taken from university of Hawaii (https://uhslc.soest.hawaii.edu/data/)sea level center.The data spans from 2001 to 2024, with sea level values ranging widely due to significant outliers (e.g., -32,767). The core range of realistic measurements falls between 691 and 844 (25th to 75th percentile). Preprocessing steps such as handling outliers, renaming columns for clarity, and combining date columns into a single datetime format are essential.
- The image is from Aqua MODIS Sea Surface Temperature(https://oceancolor.gsfc.nasa.gov/l3/) data for October 22, 2024, showing global sea surface temperatures during the daytime. The color-coded map represents temperature variations, with warmer waters in red and cooler waters in blue, providing valuable information for climate and oceanography studies. This data is part of the L3 dataset and is used to monitor oceanic conditions globally.

Dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8323 entries, 0 to 8322
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   2001    8323 non-null   int64
 1   12      8323 non-null   int64
 2   17      8323 non-null   int64
 3   652     8323 non-null   int64
dtypes: int64(4)
memory usage: 260.2 KB
```

Dataset


Aqua-MODIS Sea Surface Temperature (11 μ daytime)
Tue, 22 Oct 2024 (L3) (Daily)

Conclusion

- The literature emphasizes the success of deep learning models like LSTM for time-series forecasting, but challenges related to seasonality, cyclic patterns, and long-term dependencies remain.
- The proposed model combines LSTM (or GRU) with CNN, integrating both temporal learning and spatial feature extraction.
- This hybrid approach aims to improve prediction accuracy by capturing long-term trends, seasonal variations, and spatial dependencies in sea level data.
- The key objectives are to enhance prediction reliability, address cyclic patterns, and provide accurate real-time forecasts.
- By combining LSTM/GRU with CNN, the system improves sea level rise predictions for climate impact assessments and infrastructure planning, offering higher accuracy and computational efficiency.

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